

A geometric bound on the L^2 non-radiality: the shell-variance of the heat kernel

Working note for the “Keep the Angle” program

Purpose. The distributional master inequality (companion note “green-rank-positivity”, Theorem “Distributional master inequality”) gives $\rho_G(M) \geq 1 - 18(\phi_\Psi^2 V_\Psi)^{1/3}$, reducing Green-rank positivity to a bound on the L^2 non-radiality $V_\Psi = \mathbb{E}[(-G - \Psi \circ d_M)^2]$. That note left V as a quantity to be “bounded by heat-kernel anisotropy” without proof. This note supplies the reduction. Its three rigorous outputs: (i) the optimal V is exactly the *within-distance-shell variance of the Green kernel*, $V_{\min} = \mathbb{E}[\text{Var}(G \mid d_M)]$; (ii) via the heat representation it equals the integrated *shell-variance of the heat kernel*, which *vanishes identically on two-point homogeneous spaces* and is dominated by a heat-trace envelope whose integral converges *exactly* for $d \leq 3$; and (iii) an explicit large-time spectral-gap tail plus a quantitative perturbative bound $V_{\min} \leq C(g_0)\|g - g_0\|_{C^2}^2$ near any two-point homogeneous g_0 . The one non-explicit constant—the small-time curvature-anisotropy amplitude—is isolated and named. Notation follows the companion notes; (X, Y) are i.i.d. from the unit-volume measure and all L^2 norms are over $M \times M$ with that measure.

1 The optimal profile is the distance-conditional mean

Proposition 1 (Conditional-variance characterization). *Over all (Borel) radial profiles Ψ , $V_\Psi = \mathbb{E}[(-G(X, Y) - \Psi(d_M(X, Y)))^2]$ is minimized by the distance-conditional mean $\Psi^*(r) = -\mathbb{E}[G(X, Y) \mid d_M(X, Y) = r]$, and*

$$V_{\min} := V_{\Psi^*} = \mathbb{E}[\text{Var}(G(X, Y) \mid d_M(X, Y))].$$

Near the diagonal Ψ^ is strictly increasing with $\phi_{\Psi^*} < \infty$; its global admissibility is condition (A^*) below.*

Proof. For fixed r the map $c \mapsto \mathbb{E}[(-G - c)^2 \mid d_M = r]$ is minimized at $c = \mathbb{E}[-G \mid d_M = r]$ with minimal value $\text{Var}(-G \mid d_M = r) = \text{Var}(G \mid d_M = r)$; averaging over the law of $d_M(X, Y)$ gives the claim (the standard L^2 -projection onto $\sigma(d_M)$). For the near-diagonal regularity: writing $G = \gamma_d \circ d_M + h$ (parametrix lemma, companion note), $\mathbb{E}[G \mid d_M = r] = \gamma_d(r) + \mathbb{E}[h \mid d_M = r]$ with h bounded, so $\Psi^* = -\gamma_d - \mathbb{E}[h \mid d_M = \cdot]$ has the strictly *increasing* singular part $-\gamma_d$ (as γ_d decreases in r) dominating a bounded correction; the $\phi_{\Psi^*} < \infty$ computation is then the parametrix change-of-variables verbatim. $\square$

Remark 2 (The admissibility condition (A^*)). The bound below controls the *unconstrained* minimizer $V_{\min}$, whose profile Ψ^* must itself be a monotone profile with finite ϕ before $V_{\min}$ may enter the distributional master inequality. We isolate this as (A^*) $r \mapsto \mathbb{E}[G \mid d_M = r]$ *is strictly decreasing on $(0, \text{diam } M)$, with $\phi_{\Psi^*} < \infty$* . By the Proposition, (A^*) holds near the diagonal unconditionally; and it holds on a C^2 -neighborhood of any two-point homogeneous

metric, where $\mathbb{E}[G \mid d_M] = g(d_M)$ is strictly decreasing (Paper I) and strict monotonicity plus $\phi < \infty$ persist under small C^2 perturbation. It is a checkable, measurable condition (evaluate the shell-mean of G). Away from these regimes it is a hypothesis; where it fails one falls back on the fixed parametrix profile at the cost of losing the vanishing on two-point homogeneous spaces.

2 Heat representation: the shell-variance of the heat kernel

Let $p_t(x, y) = \sum_{k \geq 0} e^{-\mu_k t} \varphi_k(x) \varphi_k(y)$ be the heat kernel ($\mu_0 = 0$, $\varphi_0 \equiv 1$), so $G(x, y) = \int_0^\infty (p_t(x, y) - 1) dt$. Write the shell-centering $P_{\text{rad}} F(x, y) = \mathbb{E}[F \mid d_M] = F$'s conditional mean given distance, and $\delta_t = p_t - P_{\text{rad}} p_t$ (the part of the time- t heat kernel that is *not* a function of distance). Define the *time- t heat-kernel non-radiality*

$$\mathcal{N}(t) := \|\delta_t\|_{L^2}^2 = (\mathbb{E}[\text{Var}(p_t(X, Y) \mid d_M(X, Y))])^{1/2}.$$

Lemma 3 (Reduction to the heat-kernel non-radiality).

$$V_{\min}^{1/2} \leq \int_0^\infty \mathcal{N}(t) dt, \quad \text{and} \quad \mathcal{N}(t) \leq (Z(2t) - 1)^{1/2},$$

where $Z(s) = \text{Tr } e^{-s\Delta} = \sum_{k \geq 0} e^{-\mu_k s}$ is the heat trace. Consequently $\mathcal{N}(t) \equiv 0$ when M is two-point homogeneous (there p_t is a function of distance for every t), and the envelope integral $\int_0^\infty (Z(2t) - 1)^{1/2} dt$ converges iff $d \leq 3$.

Proof. The optimal remainder is $R^* = -G - \Psi^* \circ d_M = -(G - \mathbb{E}[G \mid d_M]) = -\int_0^\infty (p_t - P_{\text{rad}} p_t) dt = -\int_0^\infty \delta_t dt$, the interchange of $\int dt$ and the (bounded, linear) conditional expectation being justified by absolute integrability of $p_t - 1$ in L^2 (below). Minkowski's integral inequality gives $V_{\min}^{1/2} = \|R^*\|_2 \leq \int_0^\infty \|\delta_t\|_2 dt = \int_0^\infty \mathcal{N}(t) dt$.

For the envelope: shell-centering is an L^2 -orthogonal projection, hence a contraction, so by the law of total variance $\mathcal{N}(t)^2 = \mathbb{E}[\text{Var}(p_t \mid d_M)] \leq \text{Var}(p_t(X, Y)) = \mathbb{E}[p_t^2] - 1 = \|p_t\|_2^2 - 1 = \sum_{k \geq 0} e^{-2\mu_k t} - 1 = Z(2t) - 1$ (using $\mathbb{E}[p_t(X, Y)] = \iint p_t = 1$ and Parseval). Two-point homogeneity makes the invariant kernel p_t a function of geodesic distance, so $\delta_t \equiv 0$ and $\mathcal{N} \equiv 0$. Finally $Z(2t) - 1 = \sum_{k \geq 1} e^{-2\mu_k t}$: as $t \rightarrow 0$, $Z(2t) \sim (8\pi t)^{-d/2}$ (Weyl heat-trace asymptotics, $\text{vol} = 1$), so $(Z(2t) - 1)^{1/2} \sim c t^{-d/4}$, integrable at 0 iff $d/4 < 1$, i.e. $d \leq 3$; as $t \rightarrow \infty$, $Z(2t) - 1 \leq C e^{-2\mu_1 t}$, integrable. Hence the envelope integral converges iff $d \leq 3$, and it dominates $\int \mathcal{N}$, which is therefore finite for $d \leq 3$ (justifying the interchange above). $\square$

The critical dimension $d = 4$ reappears exactly: the shell-variance envelope ceases to be integrable, matching the Hilbert–Schmidt threshold and the fractional-filter remedy.

3 The explicit large-time tail and the small-time anisotropy

Split $\int_0^\infty \mathcal{N} = \int_0^{t_0} \mathcal{N} + \int_{t_0}^\infty \mathcal{N}$. The tail is fully explicit; the head is reduced to a named curvature-anisotropy amplitude.

Lemma 4 (Explicit spectral-gap tail). *For any $t_0 > 0$, $\int_{t_0}^\infty \mathcal{N}(t) dt \leq \frac{(Z(2t_0) - 1)^{1/2}}{\mu_1}$. Under $\text{Ric} \geq -(d-1)K$ and $\text{diam} \leq D$ the spectral gap obeys $\mu_1 \geq \mu_1(K, D) > 0$ (Li–Yau/Zhong–Yang), and $Z(2t_0) - 1 \leq C(d, K, D)$ for $t_0 \geq 1$; so the tail is bounded by an explicit function of (d, K, D, t_0) decaying like $e^{-\mu_1 t_0} / \mu_1$.*

Proof. For $t \geq t_0$, $Z(2t) - 1 = \sum_{k \geq 1} e^{-2\mu_k t} \leq e^{-2\mu_1(t-t_0)} \sum_{k \geq 1} e^{-2\mu_k t_0} = e^{-2\mu_1(t-t_0)} (Z(2t_0) - 1)$, so $\mathcal{N}(t) \leq (Z(2t_0) - 1)^{1/2} e^{-\mu_1(t-t_0)}$ by Lemma 3; integrating $\int_{t_0}^{\infty} e^{-\mu_1(t-t_0)} dt = 1/\mu_1$ gives the bound. The gap and heat-trace bounds are standard under the stated curvature/diameter hypotheses. $\square$

Definition 5 (Curvature-anisotropy amplitude). Let $\mathcal{A} := \sup_{0 < t \leq t_0} t^{d/4} \mathcal{N}(t)$, the supremal shell-standard-deviation of the heat kernel on the head interval, normalized by the diagonal Weyl scale $t^{-d/4}$.

Lemma 6 (Head bound). For $d \leq 3$, $\int_0^{t_0} \mathcal{N}(t) dt \leq \mathcal{A} \int_0^{t_0} t^{-d/4} dt = \frac{4\mathcal{A}}{4-d} t_0^{(4-d)/4}$. Here $\mathcal{A} < \infty$ always (since $\mathcal{N}(t) \leq (Z(2t) - 1)^{1/2}$, so $t^{d/4} \mathcal{N}(t)$ is bounded as $t \rightarrow 0$) and $\mathcal{A} = 0$ on two-point homogeneous spaces (there $\mathcal{N} \equiv 0$). Its small-time part $\lim_{t \rightarrow 0} t^{d/4} \mathcal{N}(t)$ is O of the shell-oscillation of the Minakshisundaram–Pleijel amplitude $u_0(x, y) = \det(d \exp)^{-1/2}$ —the variation of the Jacobi/curvature data along equal-length geodesics, $O(\sup \|\nabla \text{Ric}\|$ and the sectional-curvature variation), vanishing when the curvature is isotropic and parallel (the locally symmetric case). The full $\sup \mathcal{A}$ over $(0, t_0]$ is the manifold’s genuine heat-kernel non-radiality on that interval—generically $\Theta(1)$ for a non-symmetric metric; making it curvature-explicit uniformly in $t \leq t_0$ is the frontier item, not claimed here.

Proof. The integral bound is Definition 5 and $\int_0^{t_0} t^{-d/4} dt = \frac{4}{4-d} t_0^{(4-d)/4}$ (finite for $d < 4$). For the amplitude: near the diagonal the MP expansion $p_t(x, y) = (4\pi t)^{-d/2} e^{-d_M(x, y)^2/4t} (u_0(x, y) + u_1(x, y)t + \dots)$ has a Gaussian factor that is radial; the non-radial part of p_t at leading order is $(4\pi t)^{-d/2} e^{-d_M^2/4t} (u_0(x, y) - \mathbb{E}[u_0 \mid d_M])$, whose L^2 -shell-variance scaled by $t^{d/4}$ is O (shell-oscillation of u_0); $u_0 = \det(d \exp_x(y))^{-1/2}$ is a function of the geometry along the x -to- y geodesic, constant on distance shells exactly when the geometry is isotropic and parallel, and its shell-oscillation is $O(d(x, y)^2)$ times the curvature variation by the Jacobi-field (Van Vleck–Morette) expansion $u_0 = 1 + \frac{1}{12} \text{Ric}(\dot{\gamma}, \dot{\gamma}) r^2 + O(r^3)$ (the volume density is $\theta = \sqrt{\det g} = 1 - \frac{1}{6} \text{Ric}(\dot{\gamma}, \dot{\gamma}) r^2 + O(r^3)$, and $u_0 = \theta^{-1/2} = (\det g)^{-1/4}$). On a two-point homogeneous space u_0 is a function of distance, so $\mathcal{A} = 0$; the intermediate-time ($t \leq t_0$) contribution is likewise 0 there since $\mathcal{N} \equiv 0$. $\square$

Theorem 7 (Geometric L^2 non-radiality bound). Let M be closed of dimension $d \leq 3$, $\text{vol}(M) = 1$, $\text{Ric} \geq -(d-1)K$, $\text{diam} \leq D$, injectivity radius $\geq i_0$. Then for every $t_0 \geq 1$,

$$V_{\min}^{1/2} \leq \frac{4\mathcal{A}}{4-d} t_0^{(4-d)/4} + \frac{(Z(2t_0) - 1)^{1/2}}{\mu_1},$$

with $\mu_1 \geq \mu_1(K, D) > 0$ and $Z(2t_0) - 1 \leq C(d, K, D)$ explicit, and $\mathcal{A}$ the curvature-anisotropy amplitude of Definition 5. Consequently:

- (i) $V_{\min} = 0$ on every two-point homogeneous space ($\mathcal{A} = 0$ and $\mathcal{N} \equiv 0$), recovering $\rho_G = 1$ there;
- (ii) under (A^*) (Remark 2), combined with the distributional master inequality, $\rho_G(M) \geq 1 - 18(\phi_{\Psi^*}^2 V_{\min})^{1/3}$, so $\rho_G(M) > 0$ whenever the right side of the displayed bound is small enough that $\phi_{\Psi^*}^2 V_{\min} < 18^{-3}$ —a class explicit in (K, D, i_0, μ_1) modulo the single anisotropy amplitude $\mathcal{A}$.

Proof. Sum Lemmas 6 and 4 into Lemma 3. (i) is $\mathcal{A} = 0$ and $\mathcal{N} \equiv 0$; (ii) is the companion distributional inequality applied to Ψ^* , which is a monotone profile with $\phi_{\Psi^*} < \infty$ exactly under (A^*) . $\square$

Corollary 8 (Perturbative $\eta^{2/3}$ rate). *Let g_0 be a two-point homogeneous metric ($\text{vol} = 1$) and g a metric with $\|g - g_0\|_{C^2} \leq \eta$ small. Then $V_{\min}(g) \leq C(g_0)\eta^2$, with $C(g_0)$ the standard parabolic-perturbation constant of g_0 , and hence $\rho_G(g) \geq 1 - 18(\phi_0^2 C(g_0))^{1/3}\eta^{2/3} \rightarrow 1$ as $\eta \rightarrow 0$, quantifying the perturbative openness theorem in the L^2 index.*

Proof. By Remark 2, (A^*) holds for g near g_0 , so $V_{\min}(g)$ is admissible. Since g_0 is two-point homogeneous, $\delta_t^{g_0} \equiv 0$, so $\mathcal{N}^g(t) = \|\delta_t^g\|_2 = \|(I - P_{\text{rad}}^g)p_t^g - (I - P_{\text{rad}}^{g_0})p_t^{g_0}\|_2$; adding and subtracting,

$$\mathcal{N}^g(t) \leq \underbrace{\|p_t^g - p_t^{g_0}\|_2}_{\text{(I)}} + \underbrace{\|(P_{\text{rad}}^g - P_{\text{rad}}^{g_0})p_t^{g_0}\|_2}_{\text{(II)}}.$$

For (II): $\|d^g - d^{g_0}\|_\infty = O(\eta)$ (the metric is C^0 - $O(\eta)$ -close), so the shell-centering conditional expectations differ by $O(\eta)$ applied to the radial-regular $p_t^{g_0}$, giving (II) $\leq C\eta(Z^{g_0}(2t) - 1)^{1/2}$. For (I): the parabolic (Duhamel) perturbation estimate— $p_t^g - p_t^{g_0} = -\int_0^t p_{t-s}^g(\Delta_g - \Delta_{g_0})p_s^{g_0} ds$ with $\Delta_g - \Delta_{g_0}$ of order two and coefficients $O(\|g - g_0\|_{C^1})$ —gives (I) $\leq C\eta(Z^{g_0}(ct) - 1)^{1/2}$ (imported fact 6). Both envelopes are $\asymp t^{-d/4}$ as $t \rightarrow 0$ (integrable for $d \leq 3$) and decay like $e^{-c\mu_1(g_0)t}$ as $t \rightarrow \infty$; hence $V_{\min}(g)^{1/2} = \int_0^\infty \mathcal{N}^g(t) dt \leq C(g_0)^{1/2}\eta$. This uses only the leading Duhamel bound—not smoothness of the higher heat coefficients u_j , which $\|g - g_0\|_{C^2}$ does not control. $\square$

4 Imported facts and the frontier

Imported. (1) Weyl heat-trace asymptotics $Z(s) \sim (4\pi s)^{-d/2} \text{vol}$ as $s \rightarrow 0$ (Minakshisundaram–Pleijel). (2) Spectral-gap lower bound $\mu_1 \geq \mu_1(K, D)$ under $\text{Ric} \geq -(d-1)K$, $\text{diam} \leq D$ (Li–Yau; Zhong–Yang for the $K = 0$ sharp form). (3) The MP short-time expansion with $u_0 = \det(d\exp)^{-1/2} = 1 + \frac{1}{12}\text{Ric}(\dot{\gamma}, \dot{\gamma})r^2 + O(r^3)$. (4) C^1 -in- C^2 continuity of the heat semigroup in the metric on compact time-intervals (Kato/parabolic perturbation). (5) The distributional master inequality and the parametrix lemma (companion note “green-rank-positivity”). (6) The Duhamel heat-kernel perturbation estimate $\|p_t^g - p_t^{g_0}\|_{L^2} \leq C\|g - g_0\|_{C^1}(Z^{g_0}(ct) - 1)^{1/2}$, from $p_t^g - p_t^{g_0} = -\int_0^t p_{t-s}^g(\Delta_g - \Delta_{g_0})p_s^{g_0} ds$ and the $L^2 \rightarrow L^2$ bounds on the semigroup (standard parabolic perturbation).

What is now proved. The L^2 non-radiality is the shell-variance of the Green kernel, equals the integrated heat-kernel shell-variance, *vanishes on two-point homogeneous spaces*, is finite for $d \leq 3$ (envelope integrable exactly there), has an explicit spectral-gap tail, and is $O(\eta^2)$ within a C^2 -neighborhood of any symmetric metric—so $\rho_G \geq 1 - C(g_0)\eta^{2/3} \rightarrow 1$, a quantitative openness (the earlier openness theorem was purely qualitative). The empirical plateau ($\rho_G \gtrsim 0.79$ across the hunt families, **experiments/green-rank**) is the statement that $\mathcal{A}$ and $V_{\min}$ stay bounded even for strong bottlenecks.

Frontier. Making the anisotropy amplitude $\mathcal{A}$ explicit in $(K, D, i_0, \sup\|\nabla \text{Ric}\|)$ —the leading-order estimate of Lemma 6 promoted to a two-sided bound uniform in $t \leq t_0$ —would upgrade Theorem 7 from a named- $\mathcal{A}$ class to a closed curvature/diameter theorem. This is the one remaining non-explicit constant; everything else in the chain is explicit.